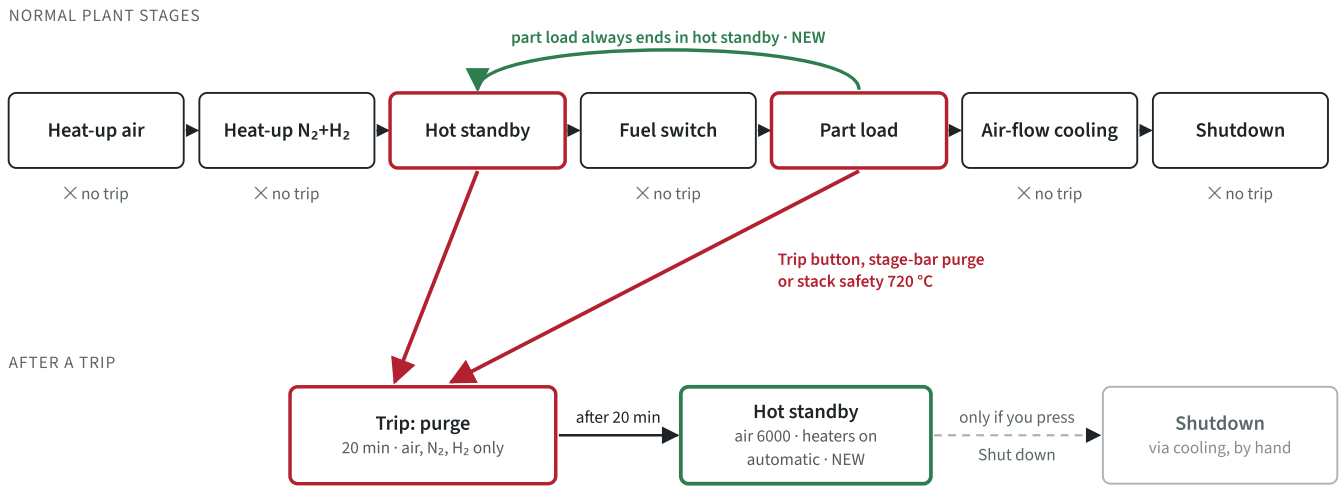
DIGITAL TWIN · 4 NOV 2026 · VERSION 15

Trip and part-load rules in the twin

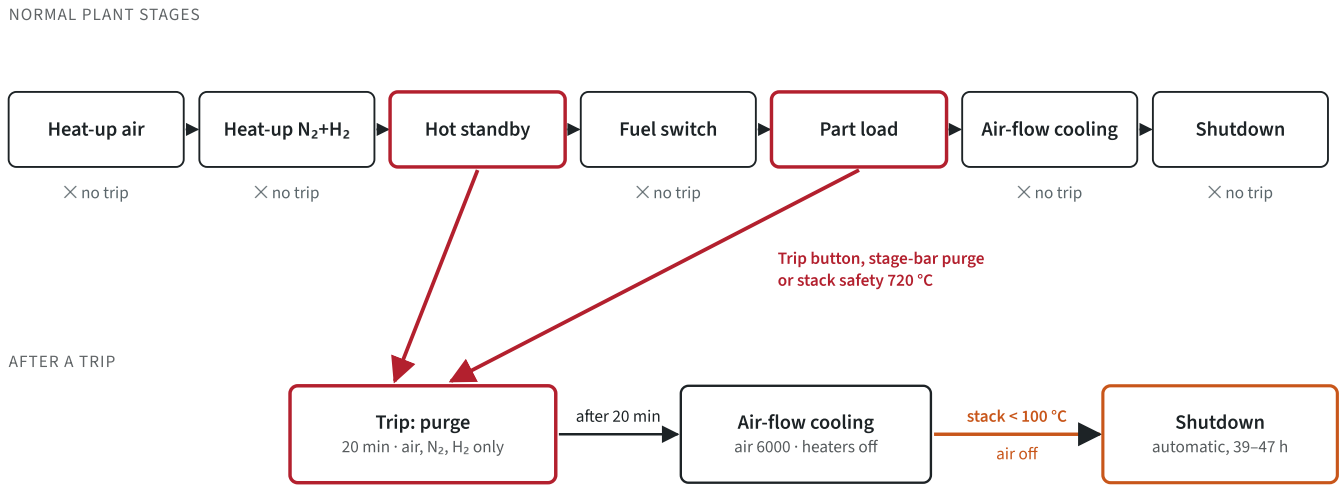
A trip purge is allowed only from **hot standby** or **part load**. After the purge the twin now goes **back to hot standby** by itself. And part load now always ends in hot standby.

A trip is the plant's emergency stop. The **purge** is its first step: gas and current are cut, and only air, nitrogen and hydrogen flow for 20 minutes to flush the fuel side. Earlier today (version 14) the twin cooled all the way down to shutdown after every trip. The plant check showed that the plant mostly went back to hot standby instead, so that rule was replaced.

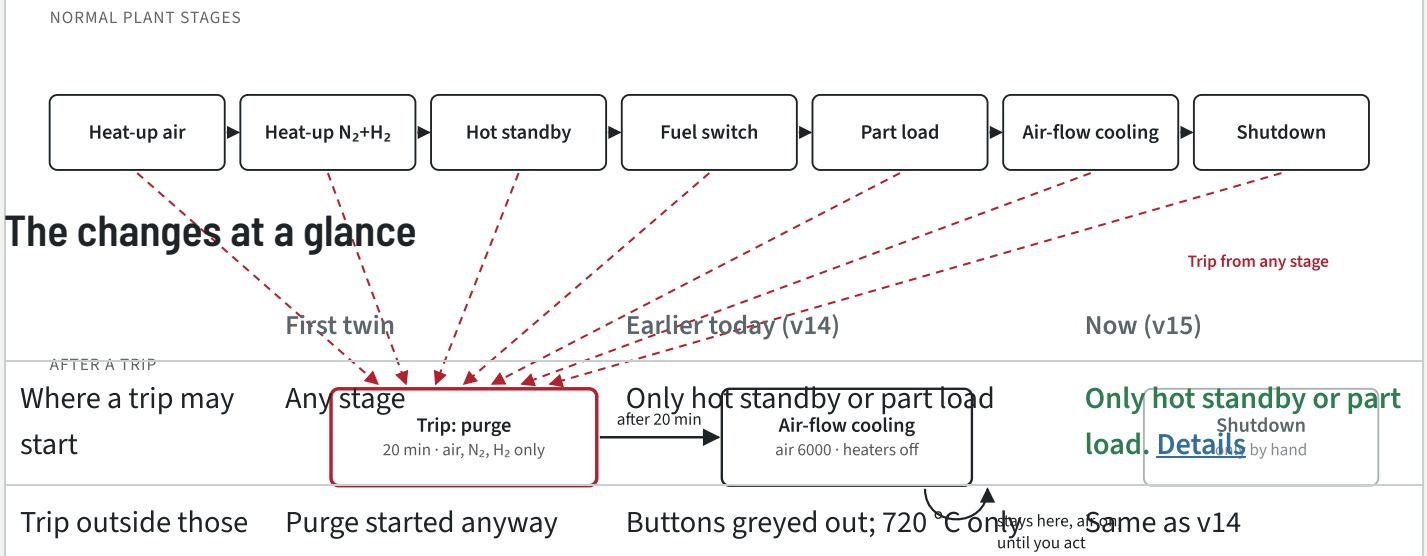
Now (v15)



Earlier today (v14)



First twin



Top row: the seven normal plant stages. Red arrows: where a trip can start. Green arc: part load always returns to hot standby. Bottom row: what follows a trip. The three drawings show now, earlier today and the first twin.

The changes at a glance

	First twin	Earlier today (v14)	Now (v15)
Where a trip may start	Any stage	Only hot standby or part load	**Only hot standby or part load. Details**
Trip outside those two stages	Purge started anyway	Buttons greyed out; 720 °C only warns	Same as v14
End of the purge	Air-flow cooling; air stays on until you act	Cooling, then air off below 100 °C; shutdown after 39–47 h	**Back to hot standby straight away. Details**
Leaving part load	To any stage	To any stage	**Always to hot standby first. Details**

All rules were checked by running the twin in a test browser. The published twin is updated at the usual link (version 15).

Against the real plant (page 5): the trip-start rule matches 5 of the 6 plant purges. Going back to hot standby matches 4 of the 6. The part-load rule is stricter than the plant, which sometimes went straight from part load to cooling.

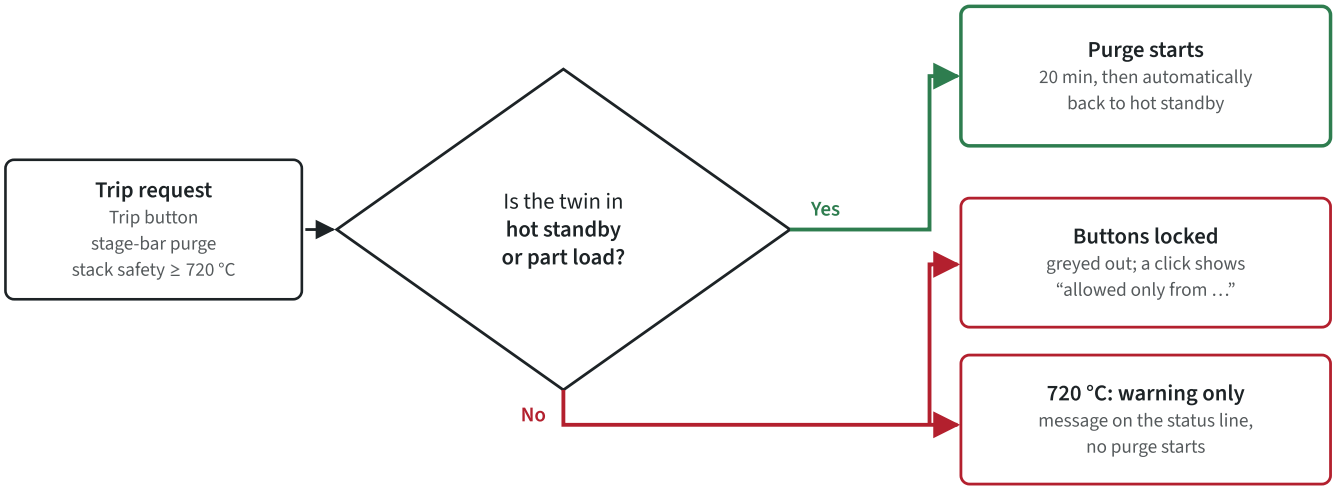
> **If you remember one thing: a trip only happens from hot standby or part load, and afterwards the twin is back in hot standby, ready to load again or to shut down.**

CHANGE 1 · UNCHANGED SINCE V14

A trip is allowed only from hot standby or part load

Every way of starting a trip in the twin first asks one question: is the plant in hot standby or part load? Only then does the purge start.

There are three ways to start a trip in the twin: the red **Trip** button, the **Trip: purge** button at the end of the stage bar, and the automatic trip when the stack safety thermocouple (TIRZ 8.1.3) reads 720 °C. All three go through the same check.



One check for all three ways of starting a trip. Yes: purge, then back to hot standby. No: nothing is stopped; you only see why.

What you see on the screen

- **In hot standby or part load** both Trip buttons look normal. A click starts the 20-minute purge and the status line says “…then back to hot standby”.
- **In any other stage** both buttons are greyed out. A click shows, for example: “Trip purge is allowed only from hot standby or part load, not from fuel switch.”
- **The stage-bar purge button** purges from where the twin is. It no longer jumps the twin to 654 °C.
- **The automatic 720 °C trip** still fires in hot standby and part load. In other stages it writes a warning on the status line but starts no purge.

Test: Trip pressed in every stage

Twin stage	Trip buttons	Result
Heat-up air	Greyed out	**No purge, message shown**
Heat-up N_2+H_2	Greyed out	**No purge, message shown**
Hot standby	Open	**Purge, then hot standby**
Fuel switch	Greyed out	**No purge, message shown**
Part load	Open	**Purge, then hot standby**
Air-flow cooling	Greyed out	**No purge, message shown**
Shutdown	Greyed out	**No purge, message shown**
Fuel switch, stack pushed to 740 °C	Automatic	Warning only, no purge
Part load, stack keeper off	Automatic	**Trips after 77 min at 697 °C model stack (safety reading 720 °C), then hot standby**

> **If you remember one thing: outside hot standby and part load, the twin will not purge. It only tells you why.**

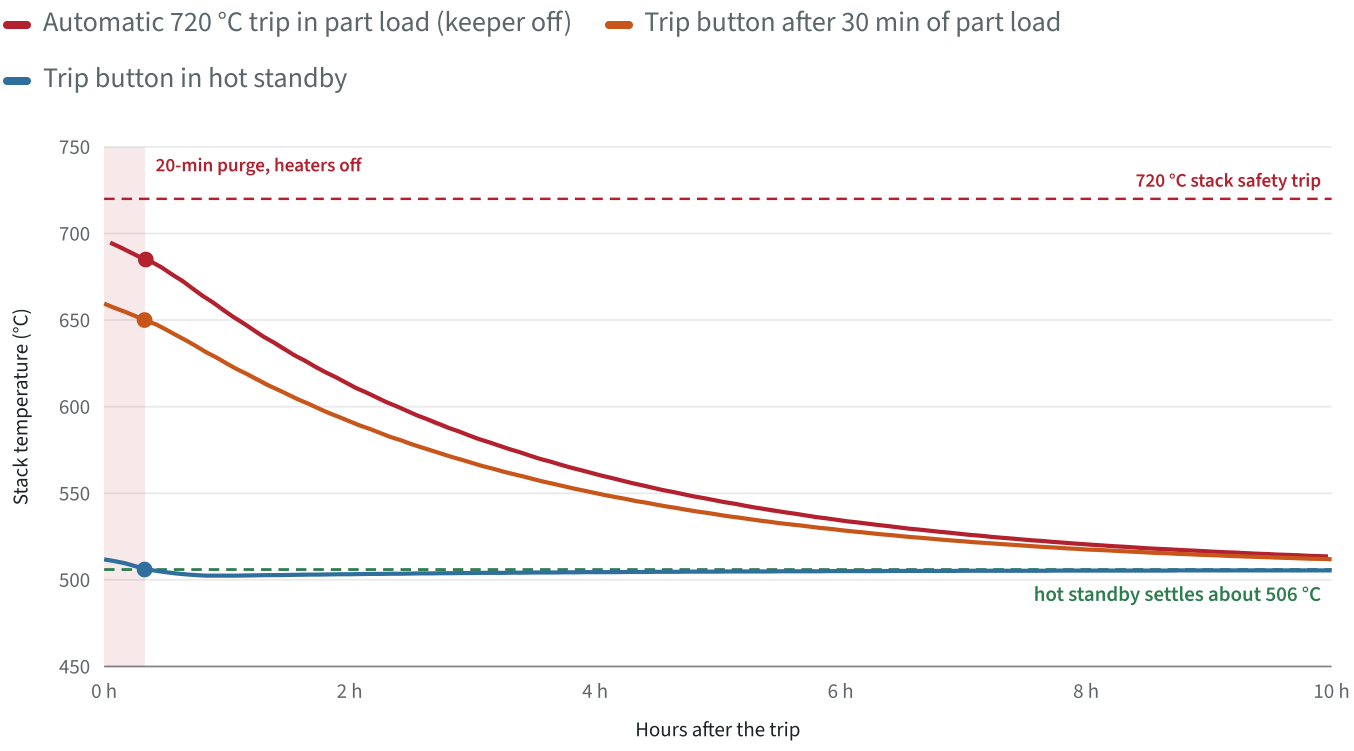
CHANGE 2 · NEW IN V15

After the purge the twin goes back to hot standby

When the 20-minute purge ends, the twin sets the hot-standby settings by itself and stays there. It no longer cools down to shutdown.

In version 14 the purge was followed by air-flow cooling and an automatic shutdown at 100 °C. In the twin that took 39 to 47 hours. The plant does not do that: after 4 of its 6 purges it was back in hot standby within 10 to 25 minutes (plant check). The twin now does the same.

Step	What the twin sets	Ends when
1 · Trip: purge	Gas, POx air (air for the fuel reformer) and current off; air 6000, N_2 1200, H_2 60 ln/min; heaters off	`20 min`
2 · Hot standby	Air 6000 ln/min, heaters on, air-heater target 500 °C, N_2 + H_2 automatic, gas and current stay off	`you press a button`
3 · Only if you want it	**Shut down** button from hot standby: cooling at 8000 ln/min, air off below 100 °C	`about 22 h`



Twin stack temperature after a trip, from the test runs. The red band is the purge. Each dot marks the moment the twin is back in hot standby. All three settle at the hot-standby level.

Trip from	Stack at the trip	Back in hot standby	4 h later	10 h later
Part load (Trip button after 30 min)	659 °C	20 min, 650 °C	550 °C	512 °C
Hot standby (Trip button)	512 °C	20 min, 506 °C	504 °C	506 °C
Part load, automatic 720 °C trip (after 77 min of part load)	697 °C	20 min, 685 °C	562 °C	514 °C

If you move a knob during the purge or afterwards, you take over by hand, like with any other playbook in the twin.

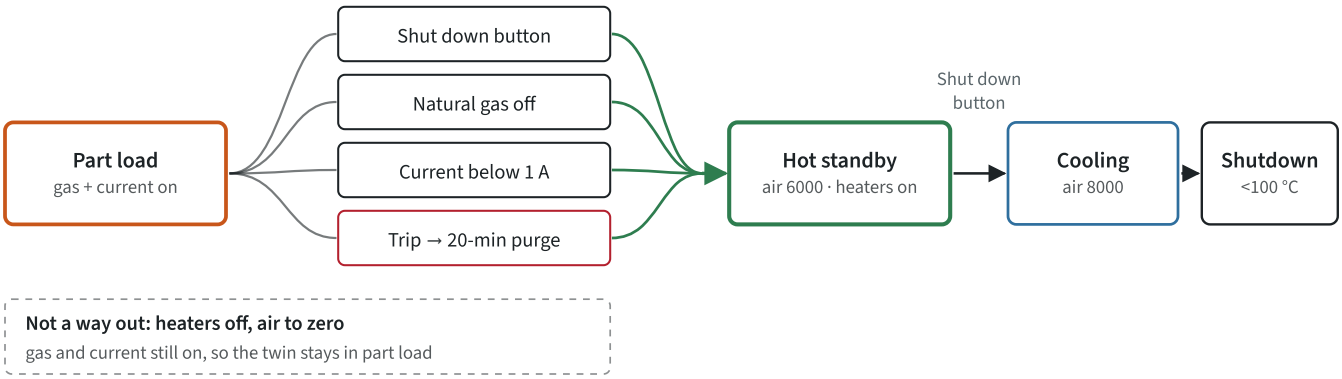
> **If you remember one thing: every trip now runs a 20-minute purge and then lands in hot standby. Shutting down is your choice, with the Shut down button.**

CHANGE 3 · NEW IN V15

Part load always ends in hot standby

Whatever takes the twin out of part load, it now unloads to hot standby first. To shut down, you press Shut down again from hot standby.

Part load is the only stage where the stack makes power: natural gas is burning and current flows. Before, pressing **Shut down** in part load sent the twin straight into cooling with the gas and load just cut. Now any way out of part load sets the hot-standby settings first: current 0, natural gas and POx air closed, N_2 + H_2 back on, air 6000 ln/min, heaters on, air-heater target 500 °C.



Every way out of part load goes to hot standby. Shutdown is reached only from hot standby. Changing heaters or air alone does not end part load.

Test in the twin

What was done in part load	Twin goes to	Note
Shut down button	**Hot standby**	Stack 649 → 561 °C after 3 h
Then Shut down again, from hot standby	Air-flow cooling → Shutdown	`22.1 h to 100 °C, air off`
Natural gas to 0	**Hot standby**	
Current to 0.5 A	**Hot standby**	Below 1 A counts as no load
Trip button	**Purge, then hot standby**	518 °C after 8 h
Heaters off	Stays part load	Gas and current still on

Air to 0	Stays part load	The stack keeper raises the air again
Stage-bar jump to air-flow cooling	Air-flow cooling	Jumps are a teleport and skip the rule
From hot standby: Go to part load	Fuel switch → part load	Unchanged: part load after 11 min

The plant itself was less strict. Of its 10 part-load runs, 2 ended in hot standby, 3 went straight to air-flow cooling, 2 to fuel switch, 1 to shutdown and 2 to a trip. The twin's rule is the safe, simple version: unload first, then decide.

> **If you remember one thing: from part load the twin always steps down to hot standby first; shutdown is one more button press away.**

CHECK AGAINST THE PLANT

Does the twin follow the real plant?

Both trip rules now mostly do. The plant purged from part load or hot standby (rule 1), and after most purges it went back to hot standby (rule 2), as the twin now does. The part-load rule is stricter than the plant.

I checked all **9 trips** in the 39 days of plant data. All of them happened in March 2026. For each trip I looked at what the plant was doing just before, how it was stopped, and what it did next. The stages and flows come from the 1-minute stage table and the 10-second sensor data.

TRIP	WHAT THE PLANT WAS DOING	WHAT HAPPENED	AFTERWARDS	RULE 1	RULE 2
#1 13 Mar	Part load · 8 A	Purge 15 min	Cooling at 6000, 10 h; log ends at 444 °C	✓	✗
#2 19 Mar	Load just ended, gas on	Purge 16 min → stop	Air off 8 min → hot standby	✗	✓
#3 19 Mar	Part load · 8 A	Purge 15 min	Cooling 14 h → re-heat at 459 °C	✓	✗
#4 24 Mar	Part load · 14 A	Purge 17 min → stop	Air off 6 min → hot standby	✓	✓
#5 24 Mar	Part load · 8 A	Purge 2 min → stop	Air off 7 min → hot standby	✓	✓
#6 25 Mar	Part load · 12 A	Emergency stop	Air off 5 min → hot standby	–	–
#7 25 Mar	Hot standby (after load)	Purge 15 min	Straight back to hot standby	✓	✓
#8 25 Mar	Cooling after load	Emergency stop	Cooling at 9000, 18 h → shutdown at 41 °C	–	–
#9 30 Mar	Start, no load yet	Emergency stop	Air off 4 min → hot standby	–	–

Red outline: emergency purge (air, N_2 and H_2 only). Dark box: emergency stop, which closes every supply, including the air, with no purge. Green outline: back to hot standby. Rule 1 (purge only from hot standby or part load) and rule 2 (back to hot standby after the purge): ✓ the twin does the same, ✗ the twin differs, – an emergency stop, which the twin does not model.

Twin against plant, point by point

Twin rule or setting	What the plant did	Match
Purge only from hot standby or part load	6 purges: 4 in part load (#1 and #3 were still carrying 8 A of current), 1 in hot standby just after load (#7), 1 a moment after load stopped with gas still on (#2). Our stage table calls #2 “fuel switch”.	Mostly: 5 of 6

After the purge: back to hot standby (new in v15)	4 of 6 purges went back to hot standby within 10–25 min. 3 of those first had an emergency stop with the air off for 6–8 min. The other 2 cooled at 6000 ln/min for 10–14 h without reaching shutdown.	**Mostly: 4 of 6**
Time to hot standby	10–25 min after the trip	Twin: 20 min, no air-off pause
Part load always ends in hot standby (new in v15)	Of 10 part-load runs: 2 → hot standby, 3 → air-flow cooling, 2 → fuel switch, 1 → shutdown, 2 → trip	Partly: the twin is stricter
Cool-down to shutdown	Only #8, an emergency stop with no purge: air 9000 ln/min, 664 → 41 °C in 18 h, then air off.	Twin: Shut down from hot standby, air 8000, about 22 h
Purge length 20 min	15–17 min (one cut short at 2 min)	Close; plant 15 min
Flows during the purge	Air 6000, N_2 1200, H_2 60 ln/min; gas, POx air and current off	**Yes**
Automatic trip at 720 °C (stack safety TIRZ 8.1.3)	The safety thermocouple read 609–696 °C at the 9 trips, never 720 °C. The trips came from other limits (logged as “power cut at trip value”) or were started by the operators.	Never happened
Air off at 100 °C	Full plant cool-downs switch the air off at 25–46 °C	**No**

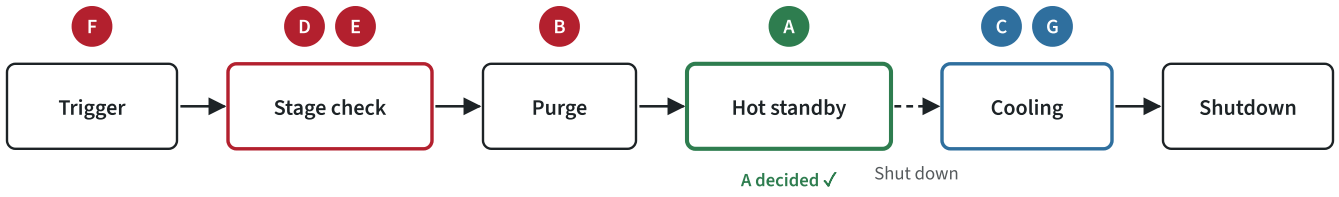
> **If you remember one thing: both trip rules now match what the plant did most of the time. The plant sometimes went straight from part load to cooling; the twin always stops in hot standby first.**

NEXT DECISIONS

What to change, if anything

Point A is now decided: after a trip the twin goes back to hot standby. The other points are still open. Nothing else has been changed.

Point	Twin now	Plant	Option
A. What follows the purge	Back to hot standby (v15)	Mostly back to hot standby (4 of 6)	Decided
B. Purge length	20 min	15–17 min	Set 15 min
C. Cool-down to shutdown	From hot standby: air 8000, off at 100 °C, about 22 h	Air 9000, off at about 40 °C, 18 h (#8)	Use 9000 ln/min and air off at 40 °C
D. Trip just after load with gas still on (#2)	Refused, because the twin calls it fuel switch	Purged	Allow a purge in fuel switch when it comes straight after part load
E. Emergency stop	Not in the twin	3 of 9 trips; 3 purges ended in a stop with the air off 6–8 min	Add an emergency-stop button: everything off, air included
F. What triggers a trip	Stack safety at 720 °C only	Never at 720 °C; other limits or the operators	Find which signal the plant's “trip value” is, then use it
G. Part load straight to cooling	Not allowed: hot standby first	3 of 10 part-load runs went straight to cooling	Keep the rule, or let Shut down in part load go straight to cooling



Where each point sits in the trip sequence. Green: decided. Red: open, around the trip itself. Blue: open, on the way to shutdown.

Where the change lives

FILES (PROJECT FOLDER TWIN_2D)

```
app.js: canPurge / startPurge: the stage check and the locked buttons
app.js: PLAYS.afterTrip + toStandby: back to hot standby after the purge
app.js: followPartLoad: part load always unloads to hot standby
index.html and README.md: the “How it works” text
```

> **If you remember one thing: A is settled. The biggest gaps left against the plant are C (cool-down at 9000 ln/min, air off near 40 °C) and E (an emergency-stop button).**